

SATYAM SHARMA

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Technical Skills

Programming Languages:	Python, Javascript, PHP, SQL.
Deep Learning & Gen AI:	RAG, LLMs, PyTorch, TensorFlow, Transformers, CNNs, AI Agents.
AI & Cloud APIs:	Hugging Face, OpenAI API, Microsoft Copilot, NVIDIA NIM.
Backend:	Node.js, GraphQL, Symfony, RESTful APIs, FastAPI, Drupal.
Frontend:	React.js, Vue.js, Twig.
Databases:	MySQL, PostgreSQL, Prisma ORM, MongoDB.
DevOps & Cloud:	Docker, Azure DevOps, CI/CD, Git, AWS (EC2, S3).
Data Analysis & Visualisations:	Pandas, NumPy, Matplotlib, PowerBI.

Work Experience

The Commonwealth Secretariat	London, United Kingdom
<i>AI Researcher, Secretary General's Office</i>	Jul 2024 - Present

– AI-Powered Regulatory Impact Assessment Tool

- Co-led the development of an AI-driven Regulatory Impact Assessment (RIA) tool, implementing **RAG** with **Python**, **LangChain** and **OpenAI API** to automate policy analysis and cost-benefit assessment.
- Optimized RAG performance by reducing **retrieved neighbors(k)**, improving retrieval precision by **~20%** while cutting API costs by **~25%**.
- Implemented secure document processing pipeline with **Node.js**, **GraphQL** and **Prisma ORM**, handling **50+** concurrent policy generation requests.
- Built responsive **Vue.js** frontend with **Apollo GraphQL**, achieving **3.2s** average page load time.
- Established containerized deployment pipeline using **Docker** and **Azure DevOps**, enabling consistent development and testing environments for future scaling.

– StrategusAI & Other Commonwealth AI Initiatives

- Integrated **vector database** capabilities into **StrategusAI** backend, implementing efficient document chunking and embedding strategies that improved search relevancy scores by **~15%**.
- Led full-stack development of Commonwealth AI Consortium website using **React.js** and **Drupal CMS**, developing **CI/CD pipelines** that reduced deployment time by **~45%** while achieving **1.2s** page loads.

<i>AI Researcher - Work Experience Program</i>	Nov 2023 - Jun 2024
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- Facilitated the fine-tuning of organization's **internal LLM**, utilizing the **LLaMA 3.1 8B model** for document analysis, enhancing the Secretariat's AI-driven insights and workflows.
- Collaborated in the development of organizational AI integration strategies, identifying **15+** automation opportunities across departments and creating implementation frameworks for ML-enabled workflows.

OBREAL Global	London, United Kingdom
<i>AI Consultant</i>	Feb 2024 - May 2024

- Evaluated **600+ AI innovation projects** for the **Global Youth Incubator Program**, assessing technical feasibility and development potential, contributing to OBREAL Global's **€2M** initiative in AI entrepreneurship.

Quilombo UK	Kingston Upon Thames, United Kingdom
<i>Full Stack Developer</i>	Jul 2023 - Oct 2023

- Enhanced Quilombo's web platform on the existing **LAMP stack** and **Symfony framework**, reducing page load times by **~20%** through PHP optimizations like caching, lazy loading, and image compression.
- Developed responsive interfaces with **Twig** templating and **bootstrap**, ensuring **98%** cross-browser compatibility and boosting mobile PageSpeed scores from **72** to **89**.

The Tann Mann Gaadi	Bengaluru, India
<i>Junior Backend Developer</i>	Jul 2021 - Jul 2022

- Optimized **PHP** and **Symfony** components within **LAMP stack** architecture, resulting in **~15%** faster page load times.
- Collaborated with senior developers to optimize **MySQL** queries and database indices, improving platform performance by **~10%**.

Education

M.Sc. - Big Data Science (*Grade: Distinction*)

London, United Kingdom

Queen Mary University of London

Sep 2022 - Sep 2023

- *Relevant Coursework:* Principles of Machine Learning, Neural Networks & Deep Learning, Cloud Computing, Statistics.
- *Dissertation:* "Advanced Machine Learning Models for Real Estate Price Prediction" (Published by CRC Press, Taylor & Francis Group)

Research & Technical Achievements

Research Publication

(CRC Press, Taylor & Francis Group 2024)

"Applications of AI for Interdisciplinary Research"

- Authored chapter on **ML models** for real estate price prediction.
- Analyzed **seven** ML algorithms for real estate valuation, reaching **89.37%** prediction accuracy with **Random Forest**.
[Paper Link] — [GitHub Repository Link]

AI Innovation Award

(**Top 15/1000+** in the **NVIDIA & LlamaIndex** Developer Contest, 2024)

"Developed ReceiptHub: AI-Powered Expense Tracker"

- Built **ReceiptHub**, an AI-based expense tracking system leveraging **NVIDIA NIM** and **Mixtral-8x7B** models.
- Implemented an end-to-end solution with **LlamaParse** for intelligent receipt parsing, enabling **automated expense categorization** and **visualization**.
- Full project implementation and documentation - [GitHub Repository Link]

Professional Certifications

- **Microsoft Certified:** Azure Data Engineer Associate
- **Microsoft Certified:** Azure Data Analyst Associate
- **Google** Data Analytics Professional Certificate